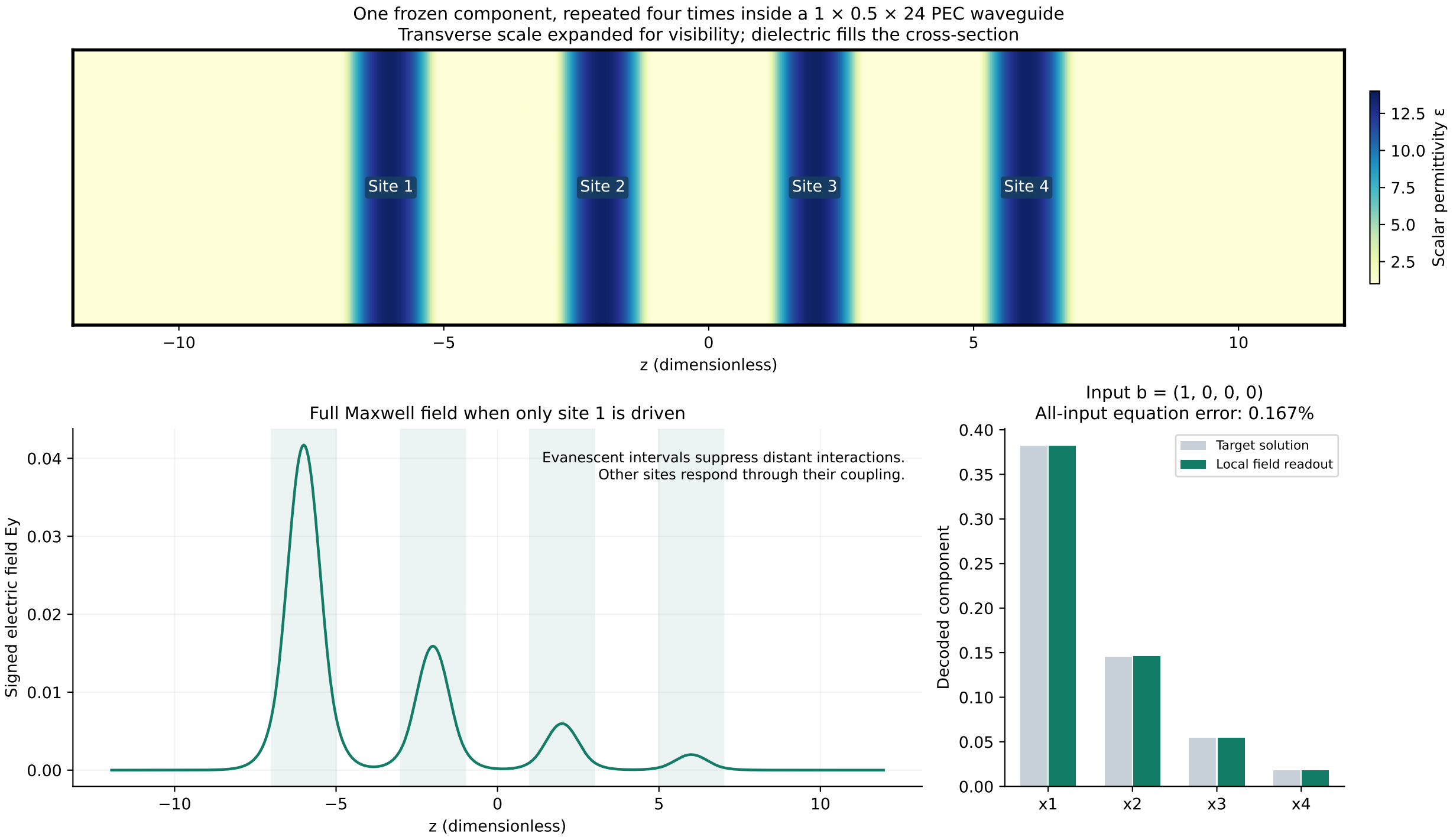
One frozen component, repeated four times inside a 1 × 0.5 × 24 PEC waveguide
Transverse scale expanded for visibility; dielectric fills the cross-section
Site 1
Site 2
Site 3
Site 4
Scalar permittivity ε
12.5
10.0
7.5
5.0
2.5
−10
−5
0
5
10
z (dimensionless)
Full Maxwell field when only site 1 is driven
Evanescent intervals suppress distant interactions.
Other sites respond through their coupling.
Signed electric field Ey
0.04
0.03
0.02
0.01
0.00
−10
−5
0
5
10
z (dimensionless)
Input b = (1, 0, 0, 0)
All-input equation error: 0.167%
Target solution
Local field readout
Decoded component
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
x1
x2
x3
x4